

This is the Accepted Manuscript version of an article published by Taylor & Francis in the Journal *Planning Practice & Research* in 2026, which is available at: <https://doi.org/10.1080/02697459.2026.2641972>.

Cite as: Tran, V.T., Nguyen, T.H.T*, Dang, K.B., Do, T.T.T, Pham, T.P., ...Doan, Q.C. (2026). Developing a framework of land types for Vietnam's national land use planning: integrating landscape ecology, regional linkage, and climate change, *Planning Practice & Research*, 1–25. <https://doi.org/10.1080/02697459.2026.2641972>.

Developing a framework of land types for Vietnam's national land use planning: integrating landscape ecology, regional linkage, and climate change

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Abstract: National Land Use Planning in Vietnam holds the supreme authority in guiding land allocation to provinces. However, its current land types often fall short in providing the necessary framework for optimizing planning outcomes. This research aims to develop a novel, comprehensive set of land types designed for national planning, structured in five main categories: Economic development, Social development, Landscape ecology - Environment, Regional linkage, and Climate change. Utilizing a rigorous two-round Delphi method, the research elicited expert insights from 67 participants initially, and 60 in the second round. The outcomes are a proposal of 50 distinct land types, providing a critical update for Vietnam's new national land use planning cycle.

Keywords: national land use planning, a framework of land types, Delphi approach

1. Introduction

Land use change has significantly impacted a vast portion of the global land area, with 32% (equivalent to 43 millions km²) affected between 1960 and 2019 alone (Winkler et al., 2021).

Decisions made today regarding land use will have lasting effects on the quality and categories of land in the future. Consequently, land use planning (LUP) is widely regarded as a key tool in land management, capable of mitigating land use conflicts and addressing economic, social, and environmental needs (Metternicht, 2017; USAID, 2021). However, global experience indicates that traditional LUP often falls short of encompassing all critical decision-making dimensions, thereby hindering the achievement of sustainability goals (Delphin et al., 2022). For instance, China's land classification system for LUP primarily focuses on agricultural and construction land, neglecting ecological considerations, which poses challenges for multi-dimensional controls (Chen et al., 2023). Similarly, a focus on construction land in Turkey has led to the encroachment upon fertile agricultural land (Cengiz, 2013), and in many African city-regions, it has resulted in the degradation of environmentally sensitive zones (Enoguanbhor et al., 2021). These examples underscore that for LUP to truly fulfill its multi-objective role, planning approaches can adopt a comprehensive and multi-faceted perspective that moves beyond narrow, traditional and classifications.

This global challenge of effectively implementing comprehensive land use planning is particularly evident in Vietnam's National Land Use Planning (NLUP). While the country's policies on NLUP explicitly mandate the integration of landscape ecology-environment, regional linkage, climate change, and sustainable development (The Government of the Socialist Republic of Vietnam, 2020; The National Assembly of the Socialist Republic of Vietnam, 2013, 2021), the current set of land types for NLUP falls short of these requirements. Vietnam's current land type system, as applied in land use planning at national and provincial levels, remains predominantly sectoral and rigid, reflecting administrative or economic management purposes rather than environmental or spatial dynamics. This system often categorizes land into major classes such as agricultural land, non-agriculture land, and unused land—each with subtypes governed by regulatory standards. However, it lacks functional categories that could support the integration of broader planning priorities such as landscape ecological corridors, multi-functional land uses, climate adaptation zones, or green-blue infrastructure. For instance, there is no land use class that accounts for ecological buffer zones, wetland ecosystems, or climate-sensitive areas. As a result, the classification system limits the ability of planners to incorporate cross-cutting issues like regional linkages (e.g., coordinating industrial corridors or ecological networks across provinces) and climate change resilience (e.g., allocating land for flood retention). As Vietnam continues its push for industrialization and modernization, it simultaneously faces a decline in agricultural land (Ministry of Natural Resources and Environment, 2014, 2024), degradation of forest ecosystems and biodiversity (Thuairé et al., 2021), and significant

risks from climate change (Espagne et al., 2021). These converging pressures present substantial challenges for NLUP in Vietnam, hindering its ability to align with national strategies and achieve practical effectiveness.

Despite the critical need for an updated NLUP framework in Vietnam, there is a notable lack of comprehensive scientific research addressing this complex, macro-scale issue. Only two studies have directly examined the issue of land types within Vietnam's NLUP. Hoang Thi Van Anh (2011) proposed a set of 10 national-level land types, but this was primarily based on the author's conceptual framework, without explicit integration of landscape ecology, regional linkage, or climate change. Later, Nguyen Dac Nhan (2019) emphasized the importance of integrating regional linkage and climate change into the national land use management, yet did not propose any specific categories or land types (Nguyen Dac Nhan, 2019).

Therefore, this paper aims to establish a comprehensive framework of land types for the integration of landscape ecology, regional linkage, and climate change into Vietnam's NLUP. Utilizing the Delphi approach, we gathered opinions from experts in land use management across the country. The main goal of this research is to enhance Vietnam's NLUP and promote sustainable development from a scientific perspective. Specifically, this study also seeks to provide valuable insights for the upcoming guiding circulars that will accompany the Land Law 2024.

2. Background and context

2.1. National Land Use Planning

NLUP is a comprehensive approach to land use management, designed to define spatial areas and allocate land use across entire regions and sectors of a country (FAO, 1993). It typically addresses issues of national significance and international linkages, providing a strategic framework for land governance (Therivel & Therivel, 2023).

Worldwide, NLUP aims to serve multiple purposes, promoting socio-economic development alongside environmental protection in both developing countries (Ministry of Environment - Natural Resources - Physical Planning and Fisheries, 2014; Ministry of Natural Resources, 2017; Ministry of Works and Human Settlement - Japan International Cooperation Agency, 2019) and developed nations (Hendrik W van der Kamp, 2023; Ministry of Land - Infrastructure - Transport and Tourism, 2017). NLUP of each country is unique, shaped by its specific geographical, socio-economic and political contexts.

For instance, countries influenced by Buddhist philosophy, such as Bhutan and Japan, emphasize collaboration, social networks, and shared economic foundations within their national land use plans. In contrast, capitalist nations like Germany, Costa Rica, and Iceland tend to prioritize individual and economic interests (Therivel & Therivel, 2023).

Furthermore, the protection of diverse agricultural and natural land areas, beyond just forests, varies by country, often depending on the identification of soil types with high carbon stocks. Examples include Iceland's focus on safeguarding wetlands, Scotland's protection of peatlands, and Germany and Iceland's efforts to preserve natural carbon sinks, particularly swampland. Switzerland prioritizes agricultural land for both carbon sequestration and food security, while Portugal safeguards greenfield land to minimize climate change risks (Therivel & Therivel, 2023).

NLUP approaches also differ in their primary objectives. China, for example, has broadened its focus from solely urban economic development to encompass multiple economic, social, and environmental goals to ensure overall holistic urban-rural growth. Consequently, China's national policies now include urban land as a fundamental land use type, even though it primarily addresses local issues (Han & Lai, 2012). Similarly, Bhutan's NLUP considers urban lands, rural lands, and areas with key urban or agricultural characteristics as primary land types. These are principally designated for "human settlement" and "land conservation and protection," encompassing sustainably managed forest land, and natural and cultural conservation areas (Ministry of Works and Human Settlement - Japan International Cooperation Agency, 2019). In Russia, NLUP establishes broader range of land categories to ensure comprehensive development across economic, social, and landscape ecology – environment dimensions, aligning with sustainable development principles (Varlamov, 1999; Volkov, 2001).

Given these diverse approaches, the land types by national governments to manage NLUP at the national level vary significantly among countries. This divergence is primarily driven by their differing priority development strategies and unique geographical conditions.

2.2. National Land Use Planning in Vietnam

In Vietnam, NLUP is officially defined as "the national-level planning that outlines the national comprehensive planning for the allocation and zoning of land for various sectors, fields, and localities based on the land potential" (The National Assembly of the Socialist Republic of Vietnam, 2017). A key function of NLUP, as stipulated by Decree 37/ND-CP, is to identify and allocate specific land use types at the national level for lower-level localities over a ten-year period (Government of

the Socialist Republic of Vietnam, 2019). These land use types are referred to as “indicators” or “land use indicators” in Vietnamese legal documents.

Vietnam’s NLUP has been formulated for three distinct periods: 2001-2010, 2011-2020, and 2021-2030. The initial planning period, 2001-2010, primarily focused on land allocation and arrangement to support the nascent stages of industrialization and modernization, consistent with the implementation of the Land Law 1993 and 2003 (The National Assembly of the Socialist Republic of Vietnam, 1993, 2003). During this period, a framework with 46 land types was specified by the Ministry of Natural Resources and Environment (2004) (Ministry of Natural Resources and Environment, 2004), applicable across all four administrative levels.

Subsequently, in response to identified shortcomings, the NLUP for the period 2011-2020 was established, incorporating revised content and land types as per Decree No. 69/2009 of the Government of Vietnam and the Land Law of 2013. This decree emphasized a comprehensive approach to NLUP, highlighting macro-level land types of strategic significance and those requiring strict protection. There were 19 land types suggested for allocation at the national level. The National Assembly later approved the adjustment of the NLUP for the 2016-2020 sub-period through Resolution No. 134/2016/QH13. This resolution introduced new land types such as economic zone land, high-tech park land, and urban land, while simultaneously removing salt production land and aquaculture land (The National Assembly of the Socialist Republic of Vietnam, 2016) (Table 1).

Table 1. A framework of land types in Vietnam’s NLUP from 2011-2030

No	Land types	Period 2011-2020		Period 2021-2030 (Resolution QH No. 39 13/11/2021)
		Resolution QH No. 17 22/11/2011	Resolution QH No. 134 9/4/2016	
1	Agricultural land			
	Includes:			
1.1	Rice cultivation land	X	X	X
	<i>Includes: Paddy land</i>	X	X	X
1.2	Protective forest land	X	X	X
1.3	Special-use forest land	X	X	X
1.4	Production forest land	X	X	
	<i>Includes: Production forest land within natural forests</i>			X
1.5	Salt production land	X		
1.6	Aquaculture land	X		
2	Non-agricultural land			
	Includes:			

No	Land types	Period 2011-2020		Period 2021-2030 (Resolution QH No. 39 13/11/2021)
		Resolution QH No. 17 22/11/2011	Resolution QH No. 134 9/4/2016	
2.1	Industrial land	X	X	X
2.2	Land for national defense	X	X	X
2.3	Land for security purposes	X	X	X
2.4	National infrastructure development land <i>Includes:</i>			
-	Land for transportation			X
-	Land for cultural facilities	X	X	X
-	Land for medical facilities	X	X	X
-	Land for Education and training facilities	X	X	X
-	Land for sports facilities	X	X	X
-	Land for energy infrastructure			X
-	Land for post and telecommunications facilities			X
2.5	Land for national warehouses			X
2.6	Land for cultural and historical relics	X	X	X
2.7	Land for waste disposal	X	X	X
2.8	Urban residential land	X	X	
3	Unused land			
3.1	Unused land put into use	X	X	X
3.2	Remaining unused land	X	X	X
4	Economic zone land		X	X
5	High-tech park land		X	X
6	Urban land		X	X
Total number of land types		19	20	23

Source: (The National Assembly of the Socialist Republic of Vietnam, 2011, 2016, 2021)

The current framework of land types in NLUP for the period 2021-2030 aims for a more balanced approach, integrating economic and social development with national defense and security, while also prioritizing landscape ecology protection, biodiversity conservation, water source management, and adaptation to climate change and rising sea levels (The Government of the Socialist Republic of Vietnam, 2020). The content of this planning is guided by Vietnam's first comprehensive planning legislation, the Planning Law of 2017. The recent approval of Resolution 39/2021/QH15 by the National Assembly has further refined land types, now including land for energy infrastructure (e.g., wind and solar power), information transmission (post and telecommunications), and national reserve warehouses. However, this latest iteration is perceived as insufficiently emphasizing landscape ecology, environmental dimensions, and climate change adaptation. Furthermore, it lacks a robust categorization of land types into specific dimensions that would foster more integrated perspectives for NLUP.

2.3. Establishing a framework of land types for national land use planning in Vietnam

2.3.1. Categorization of land types

Vietnam's Planning Law (2017) defines NLUP content as "the distribution and spatial arrangement of land to achieve socio-economic development, national defense, security, environmental conservation, natural disaster mitigation, and climate change adaptation goals, considering land capabilities and the land requirements for linking regional and provincial industries and sectors (The National Assembly of the Socialist Republic of Vietnam, 2017). Building on this legal foundation and aligning with global practices in LUP, this study identifies and categorizes land types based on three critical dimensions that Vietnam's NLUP must explicitly integrate, alongside its traditional focus on economic and social development: regional linkage, ecological landscape - environment, and climate change.

Land use planning is widely recognized as a crucial tool for balancing human development with environment protection (Metternicht, 2018). Integrating landscape ecology factors, for instance, is considered one of five fundamental actions for addressing ecosystem maintenance challenges (Dale et al., 2000). International approaches vary in their ecological emphasis. For example, NLUP in Germany, Wales, and Portugal prioritize protecting ecological networks and green infrastructure to facilitate species migration amidst global temperature rises. In contrast, Japan's plan focuses on safeguarding land types like farmland and forests to promote self-sufficiency in disaster management (Therivel & Therivel, 2023).

The concept of regional linkage in NLUP, while critical, often lacks explicit discussion in many studies, which tend to focus more on the rural-urban connections due to their administrative proximity and interactions (e.g., flows of goods, people, finances, and knowledge) (Chigbu, 2021). However, integrating rural and urban land use planning is vital for addressing land ownership issues, ensuring ecosystem services (Eppler et al., 2015) and securing the rights of rural areas within regional and national rural-urban linkages (Matthaei, 2018). In Vietnam, the amended Law No. 37 related to planning explicitly mandates that "NLUP must ensure the specificity and linkage of regions" (The National Assembly of the Socialist Republic of Vietnam, 2018). Furthermore, researchers emphasize integrating regional linkage factors in land allocation to efficiently leverage internal and external strengths, ensuring balanced land use across sectors, effectively manage land resources, protect the environment, and prevent disasters (Dang Van Bao, 2015). In the context of Vietnam, regional linkage in LUP refers to the coordination and integration of land use strategies across provincial and administrative boundaries to promote coherent spatial development, reduce duplication, and optimize

resource use at a regional scale. For example, the lack of coordinated planning among neighboring provinces has led to the uncontrolled expansion of industrial zones, inefficient transportation networks, and competition over land and water resources. Therefore, strengthening regional linkage understood as inter-provincial cooperation and strategic alignment in land allocation is essential to achieving the objectives in the national master plan and regional planning frameworks. Therefore, our proposed framework for regional linkage transcend mere urban-rural connections, also encompassing linkages for disaster prevention and climate change adaptation, and the balance between sectoral land use needs and the national land resources to promote efficient and effective land use.

Land use change, driven by human activities, is a significant contributor to climate change (Thapa, 2022). Consequently, adjusting land use patterns is considered a more effective solution for adapting to climate change than solely reducing greenhouse gas emissions (Dale, 1997; Stone, 2009). NLUP thus serves as a key component in minimizing communities' vulnerability to climate change, both presently and in the future. It achieves this by enhancing resilience and preparedness for recovery and adaptation (Bajracharya et al., 2011; Deppisch, 2018; Richardson & Otero, 2012). This crucial role has led to NLUP being widely implemented globally, including in developing countries (Richardson & Otero, 2012; UNIDSR, 2012).

2.3.2. Proposing land types for NLUP

Building upon Vietnam's existing legal frameworks for integration and international practices in NLUP, we propose a novel framework of land types for NLUP in Vietnam, structured into five main categories: economic development, social development, landscape ecological – environment, regional linkage, and climate change adaptation (Table 2).

The initial set of land types was a primary topic discussed at the conference “Developing Indicators of Land Types in National Land Use Planning (NLUP) and Regional Land Use Planning (RLUP) for Integration of Ecological Landscape, Regional Linkage, Climate Change Adaptation to Ensure Sustainable Development”, held on March 31, 2022. This conference was an outcome of our national research project (coded DTDLCN-94/21). During this conference, we achieved a consensus on the perspectives and directions for establishing a set of land types for both NLUP and RLUP tailored to Vietnam's specific context. We also received feedback to separate the “socio-economic development” aspect into “economic development” and “social development”, consequently expanding the proposed set of land type categories from an initial four to five distinct ones. While detailed opinions regarding the nomenclature of land types were also discussed, specific suggestions

for adding or removing land types in the proposed set were not heavily emphasized, as these would be elicited later during our Delphi survey.

These categories and land types will subsequently be subjected to a Delphi survey with experts in relevant fields to identify the most suitable set for implementation, as described in the following section.

Table 2. Proposed framework of land types for Vietnam's NLUP

	Economic development	Social development	Landscape Ecology - Environment	Climate Change	Regional Linkage	References
1. Agricultural land						
Rice cultivation land	E1	S1		C1		Vietnamese NLUP for 2021-2030
Paddy land	E2	S2			R1	Vietnamese NLUP for 2021-2030
Land for other annual crops	E3	S3				Vietnamese NLUP for 2001-2010
Land for cultivation of perennial trees	E4	S4			R2	Vietnamese NLUP for 2001-2010
Protective forest land			L1	C2	R3	Vietnamese NLUP for 2021-2030
Special use forest land		S5	L2	C3	R4	Vietnamese NLUP for 2021-2030
Productive forest land	E5	S6		C4		Vietnamese NLUP for 2021-2030
Natural productive forest land	E6	S7	L3			Vietnamese NLUP for 2021-2030
Salt production land	E7	S8				Vietnamese NLUP for 2001-2010, 2011-2020
Aquaculture land	E8	S9				Vietnamese NLUP for 2001-2010, 2011, 2020
2. Non-agricultural land						
Land for national defense		S10				Vietnamese NLUP for 2021-2030
Land for security purposes		S11				Vietnamese NLUP for 2021-2030

	Economic development	Social development	Landscape Ecology - Environment	Climate Change	Regional Linkage	References
Land for industrial zones	E9				R5	Vietnamese NLUP for 2021-2030
Land for export processing zones	E10					Provincial land use planning in provincial planning for 2021-2030
Land for industrial clusters	E11					Provincial land use planning in provincial planning for 2021-2030
Land for trading and services	E12				R6	Provincial land use planning in provincial planning for 2021-2030
Land for non-agricultural production and business	E13				R7	Vietnamese NLUP for 2001-2010
Land for mining activities	E14					Vietnamese NLUP for 2001-2010
Land for infrastructure development	E15	S12			R8	Vietnamese NLUP for 2021-2030
Land for national warehouses	E16					Vietnamese NLUP for 2021-2030
Land for cultural and historical relics		S13				Vietnamese NLUP for 2021-2030
Land for waste dumping and treatment			L4			Vietnamese NLUP for 2021-2030
Land with historical-cultural relics or scenic spots			L5			Vietnamese NLUP for 2001-2010
Land for the protection of landscape and other special ecosystems			L6			Russian and German land use planning
Urban residential land	E17	S14			R9	Vietnamese NLUP for 2011-2020
Rural residential land	E18	S15			R10	Vietnamese NLUP for 2001-2010
Land for religious establishments		S16				Vietnamese NLUP for 2001-2010
Land with rivers, streams, and special-use water surface			L7	C5		Vietnamese NLUP for 2001-2010

	Economic development	Social development	Landscape Ecology - Environment	Climate Change	Regional Linkage	References
Other non-agricultural land	E19					Vietnamese NLUP for 2001-2010
3. Unused land			L8	C6		Vietnamese NLUP for 2021-2030
4. Land for economic zones	E20				R11	Vietnamese NLUP for 2021-2030
5. Land for hi-tech zones	E21				R12	Vietnamese NLUP for 2021-2030
6. Urban land	E22	S17		C7	R13	Vietnamese NLUP for 2021-2030
7. Land for tourism	E23	S18		C8	R14	Russian land use planning
8. Lagoon			L9			Swedish and Russian land use planning
9. Coastal reclaimed and encroachment land			L10			Proposed
10. Coastal surface water land			L11		R15	Provincial land use planning in provincial planning for 2021-2030
11. Land at risk of flooding due to rising sea level.				C9	R16	Proposed
12. Salinity intrusion-prone land				C10	R17	Proposed
13. Drought-prone land				C11	R18	Proposed
14. Landslide-prone land				C12	R19	Proposed
15. Desertification-prone land				C13		Proposed
16. Storm-prone land				C14		Proposed
17. Flood-prone land				C15		Proposed
18. Sedimentation-prone land at river mouths				C16		Proposed

3. Methodology

3.1. The Delphi method

The Delphi method is a structured survey technique designed to systematically collect and refine expert opinions through a series of "rounds" and feedback processes. This iterative approach is

particularly effective for complex research problems, enabling the development of informed solutions (Linstone & Turoff, 2011; Vernon, 2009). Its core principle lies in combining diverse individual opinions into a robust group consensus (Alshehri et al., 2015; Hsu & Sandford, 2007). The method's utility extends to the construction and development of dimensions and land types (Abreu & Mesias, 2020; Musa et al., 2019; Rikkonen et al., 2006; Scolozzi et al., 2012).

3.2. Expert selection

The minimum number of experts for a Delphi survey typically ranges from 7 to 8 (Linstone & Turoff, 2011; Sourani & Sohail, 2015). However, for a heterogeneous group, where experts possess diverse social or professional backgrounds yet focused on a specific topic, a recommended range of 20 to 60 experts is suggested (Jeste et al., 2010).

In this study, a two-round Delphi survey was conducted from May 2022 to July 2022. The initial round engaged 67 experts, all with extensive experience in NLUP. In the second round, 60 experts continued their participation, evaluating the Delphi questionnaire. This represents an 89.5% response rate from the initial participants (see Table 3).

Table 3. Characteristics of experts through two rounds of Delphi

Experts	Round 1	Round 2
Total	67	60
Genders		
Male (%)	62.7	61.7
Female (%)	37.3	38.3
Institutions of working		
Universities (%)	47.8	46.7
Research institutes (%)	13.4	13.3
National department of land administration (%)	22.4	23.3
Local department of land administration (%)	16.4	16.7

Securing enough qualified experts in NLUP can be challenging for similar studies in Vietnam. A key strength of this research is that the team successfully conducted an extensive, nationwide collection of expert opinions. This ensured comprehensive representation from six socio-economic regions, including less developed areas like the Central Highlands and Northern Mountain regions. This broad inclusion enhances the study's robustness, as its results reflect the perspectives of a wide and diverse network of experts in the field.

3.3. Consensus criteria

To rigorously assess the level of consensus among experts, we employed several key statistical measures: the median, interquartile range (IQR), standard deviation (S.d), and variation coefficient indices.

A strong consensus on the importance of an item is indicated by a high mean value coupled with low IQR and low standard deviation. This aligns with Grotorex and Dexter (2000), who consider mean values a valid indicator of importance. Furthermore, Julsrud and Priya Uteng (2015) and Perveen et al. (2017) suggest that a standard deviation below 1 on a 5-point Likert scale signifies strong expert consensus, while Goldman et al. (2008) propose that a standard deviation greater than 1.5 indicates a lower consensus level.

Regarding IQR, Murphy et al. (1998) argue that IQR is a reliable method for determining consensus, while Rayens and Hahn (2000) suggest that an IQR of less than or equal to 20% of the measurement scale is considered a good level of consensus. Therefore, on a 5-point scale, an IQR of less than or equal to 1 indicates a suitable level of consensus.

The Coefficient of Variation (CV) is utilized to measure the consistency of expert opinions across rounds (English & Kernan, 1976; Esmaeilpoorarabi et al., 2018; He et al., 2016; Kiba-Janiak, 2016). Calculated as the ratio of the standard deviation to the mean, a CV between 0 and 0.5 is generally accepted as indicating that consensus has been achieved, allowing the process to terminate (English & Kernan, 1976). Additionally, Scheibe et al. (2002) suggest that if the percentage change in CV value between two rounds is less than 15%, stability of consensus is achieved, and the Delphi survey can be considered complete.

4. Results

4.1. Round 1

In Delphi Round 1, experts were asked to rate the importance of specific land types on a 5-point Likert scale, where 5 indicated the highest priority and 1 the least priority. The results for each category are presented below.

Economic development

Figure 1 illustrates the expert evaluation of land types within the economic dimension. Within the agricultural land category, rice cultivation land (E1), paddy land (E2), and aquaculture land (E8) received high ratings, with average scores of 4.42, 4.48, and 4.12, respectively. All exhibited standard deviations below 1, indicating strong consensus. These findings align with Vietnam's current conditions and reflect the critical challenge of food security. Data show Vietnam's rice land decreased by an average of 18,816 hectares annually between 2010 and 2020, primarily due to conversion for industrial and urban expansion (Nguyen Quang Hoc, 2023). This competition between agricultural land and non-agricultural land (including land for industry, services, and housing) is a prevalent global issue (Doan, 2024; Wehrmann, 2012). Therefore, strong state intervention is essential in Vietnam to safeguard these vital land types. In contrast, land designated for other annual crops, perennial crops, and salt production received minimal assessment values within this category, suggesting that imposing spatial limitations on them at the national scale is not advisable.

For the non-agricultural land category, land designated for infrastructure development (E15), encompassing transportation and energy, was deemed significantly important for national economic growth, achieving an average score of 4.59, with a low standard deviation of 0.78. Land for industrial (E9) and economic zones (E20) followed closely with average scores of 4.57 and 4.51, ranking second and third, respectively (Figure 1). Experts viewed these land types as vital for Vietnam's continued industrialization and modernization efforts (Central Committee of the Communist Party of Vietnam, 2022). Conversely, residential land, primarily for human habitation and ensuring suitable living environments, was perceived to have less direct economic significance. Therefore, urban residential land (E17) and rural residential land (E18) received lower assessment scores of 3.34 and 3.09, respectively.

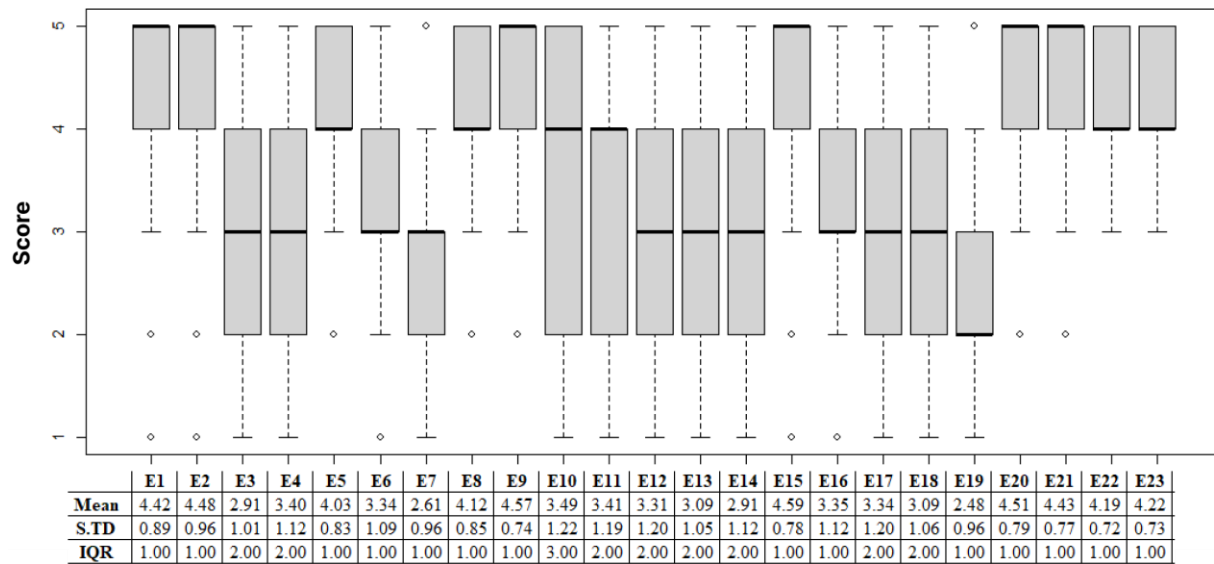


Figure 1. Results of the experts' evaluations for land types of economic development

Social development

Of the 18 land types assessed under the category of social development, 12 were deemed socially significant. Among agricultural land types, rice cultivation land (S1), paddy land (S2), and special-use forest land (S5) were evaluated as having the greatest social significance. Their average assessment scores all exceeded 4.0, with low standard deviations ranging from 0.75 to 0.82, indicating a high level of consensus on their importance in NLUP. Despite a substantial decline in the agricultural workforce over the past two decades, it still constituted 26.9% of the total working-age population in 2023, with only 42.2% of these workers possessing skilled labor (National Statistics Office, 2025). Consequently, preserving rice land not only ensures food security, as previously stated, but also sustains employment opportunities for a significant segment of the labor force.

In the non-agricultural land group, land types such as land for national defense (S10), security purposes (S11), infrastructure development (S12), and historical-cultural relics (S13), received high evaluation scores of 4.43, 4.36, 4.63, and 4.37, respectively (Figure 2). Conversely, land types related to religious establishments (S16) (including sites for churches, pagodas, communal halls, temples, shrines, and monastic retreats) received a lower score of 3.27. Expert consensus indicates that a micro-

level planning approach is more appropriate for these land types, given their closer association with local communities.

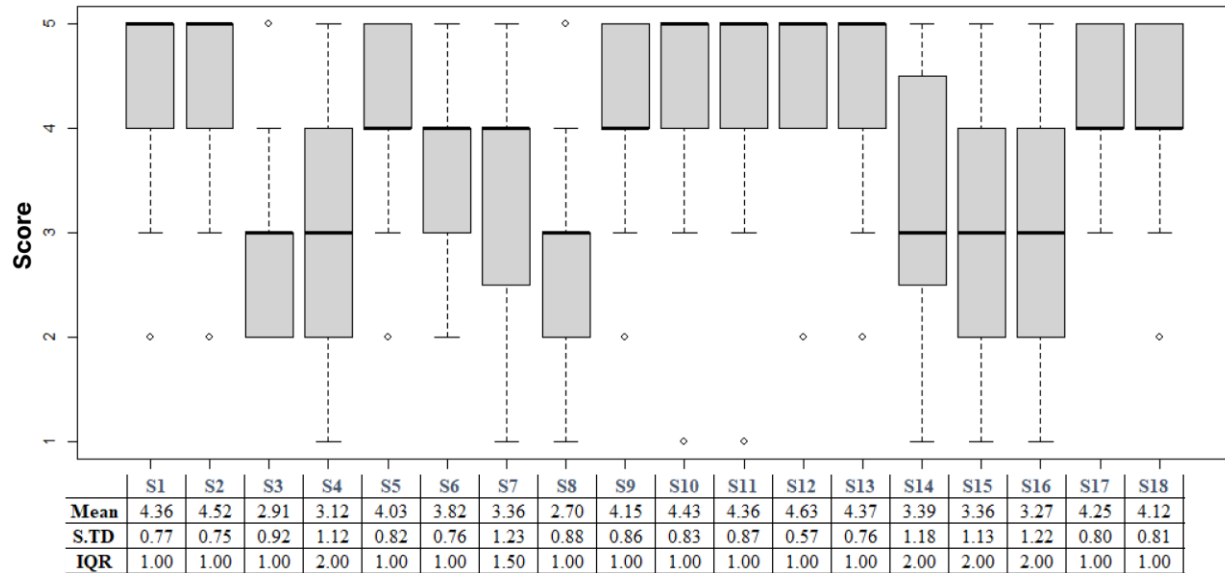


Figure 2. Results of the experts' evaluations for land types of social development

Landscape ecology – Environment

In the category of landscape ecology – environment, special-use forest land (L2) and protective forest land (L1) received the highest scores in Round 1, with means of 4.76 and 4.72, and standard deviations of 0.5 and 0.57, respectively. This finding is consistent with research by Abdul Sahim Ansari (2003), who demonstrated the significant contribution of special-use and protective forests to flood mitigation in hilly and delta regions through the reduction of flow volume. “Land with historical-cultural relics or scenic spots” (L5) despite being more related to man-made structures, was still considered important for national-level allocation by experts, significantly contributing to landscape aesthetics with a rating of 4.65. Land for the protection of landscape and other special ecosystems (L6), natural productive forest land (L3) also scored highly at 4.53 and 4.39, respectively, followed by land for waste dumping and treatment (L4) at 4.28 (Figure 3), collectively highlighting their ecological and environmental importance. Conversely, “lagoons” (L9) received relatively low ratings for national-level planning, with an average score of 3.01. This is primarily attributed to their concentration in a limited number of provinces in the Central region and their comparatively small spatial extent (approximately 50 km in length – Mai Trong Nhuan, 2007). As a result, lagoons were deemed to have insignificant areas that do not warrant national-level management.

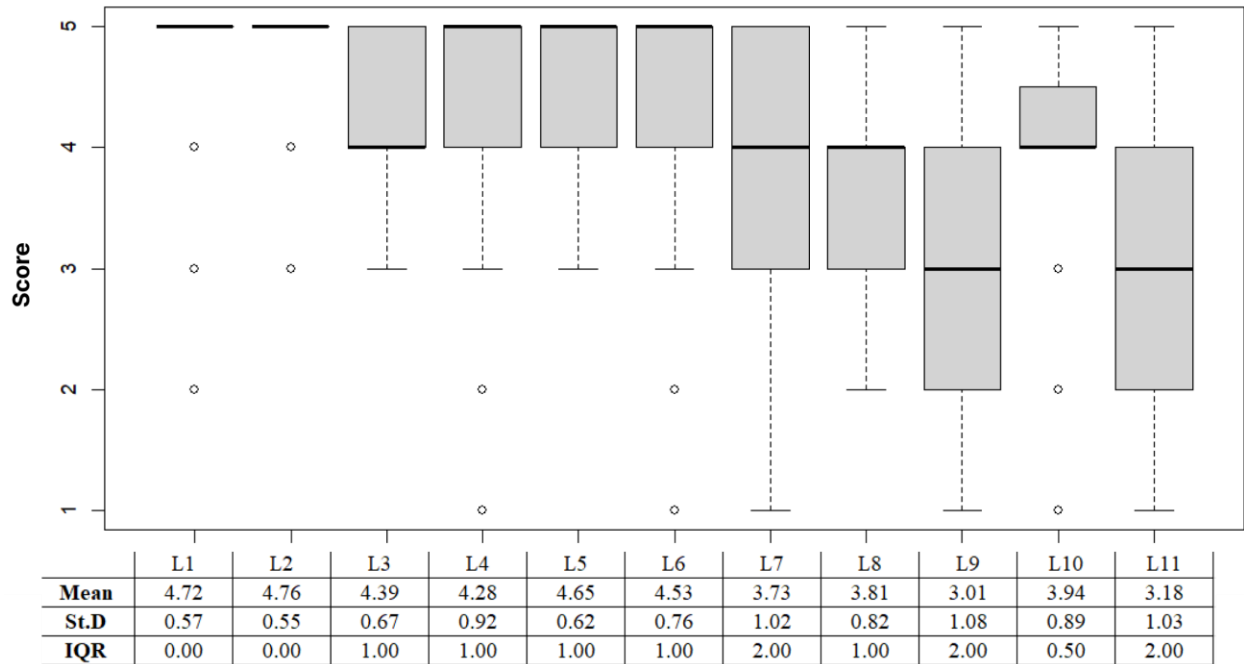


Figure 3. Results of the experts' evaluations for land types of landscape ecology - environment

Climate change

Among the 16 land types within the category of climate change, land at risk of flooding and saltwater intrusion was strongly evaluated as a necessary inclusion in the NLUP. This is highly relevant in countries considerably affected by climate change, such as Vietnam. Forecasts indicate that a 1-meter sea-level rise could submerge approximately 16.8% of the Red River Delta's area and 38.9% of the Mekong Delta's area (Ministry of Natural Resources and Environment, 2016). Furthermore, experts recommended that land types prone to drought (C11, mean = 4.38), and landslides (C12, mean = 4.42) should be explicitly considered in planning for effectively resource management and utilization. Special use forests (C3), protective forests (C2), and productive forests (C4) also received high average assessment scores of 4.3, 4.36, and 4.19, respectively (Figure 4). These evaluations align with findings from other authors (Hundecha & Bárdossy, 2004; Sun & Vose, 2016) regarding the critical role of forests in mitigating the impacts of climate change on human habitats.

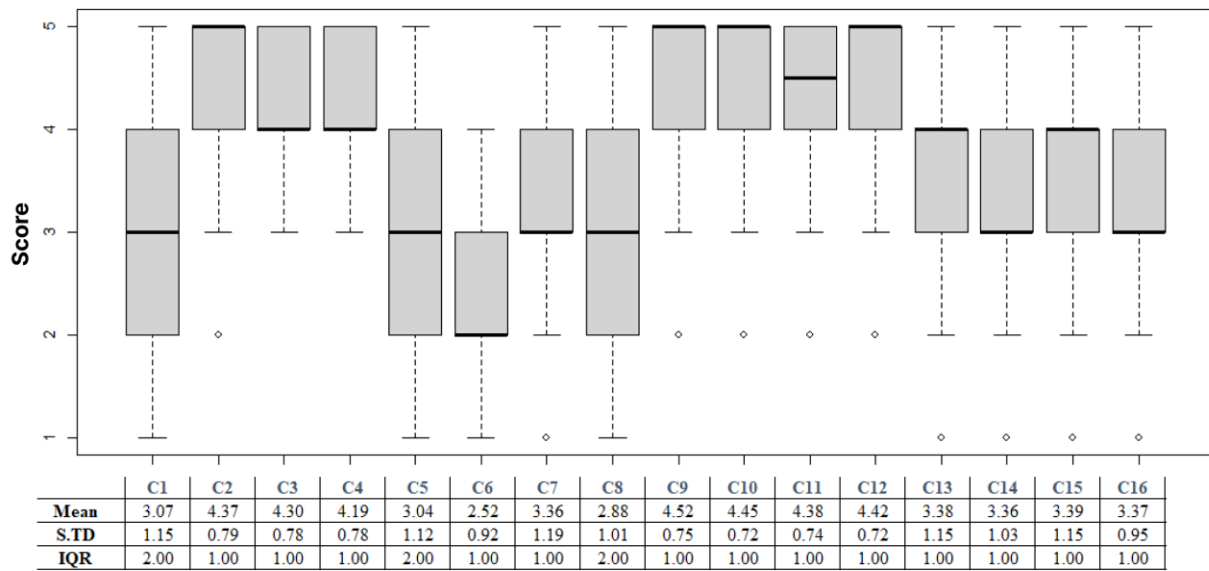


Figure 4. Results of the experts' evaluations for land types of climate change

Regional linkage

Regional linkage play a crucial role in minimizing overlapping planning and enhancing land use efficiency. However, the land types allocated for regional linkage have historically been almost absent from previous land planning documents. Out of the 19 land types used to assess in this category, 12 had an average score greater than 3.5. According to the experts, the land for infrastructure development (R8, mean = 4.46, S.d = 0.7) was considered the most important land type for ensuring regional linkage in planning. Land for industrial (R5) and economic zones (R11) ranked second and third in importance, with average scores of 4.40 and 4.31, respectively. Land types related to forest land (R3, R4), with scores greater than 4.0, were also deemed indispensable in this category. Additionally, land types affected by natural disasters and climate change were also recommended for inclusion in NLUP (Figure 5).

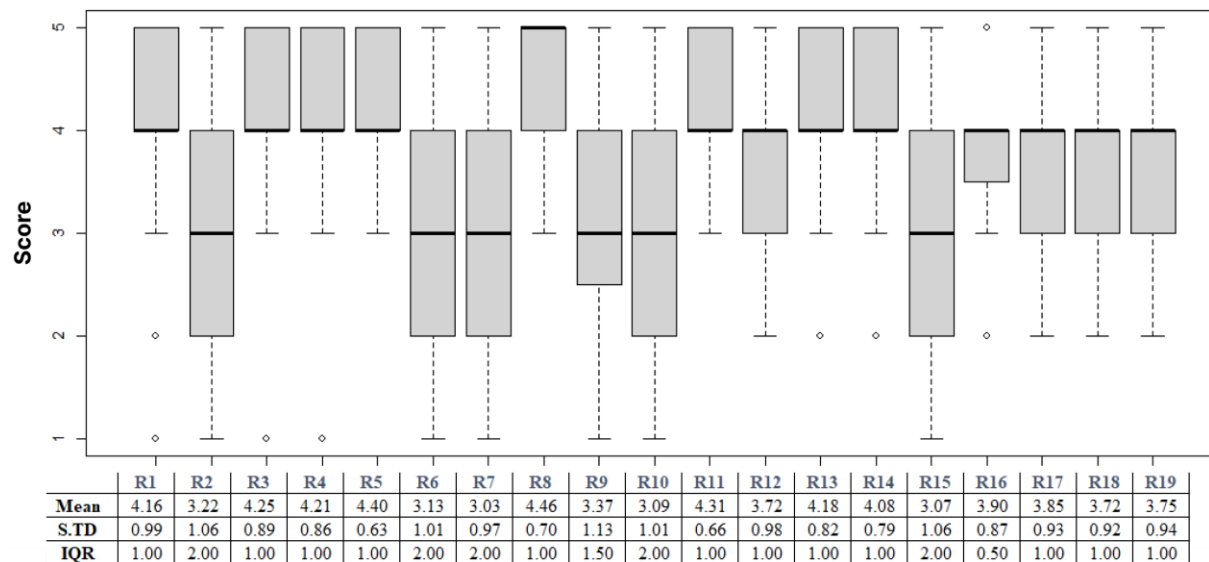


Figure 5. Results of the experts' evaluations for land types of regional linkage

The first-round Delphi revealed discrepancies in expert evaluations for certain land types. Notably, land of river, stream, and specialized water surface land (L7) within the category of ecological landscape – environment showed a lack of consensus, despite an average score above 3.5 (S.d=1.02, and IQR=2.0). This indicates a need for closer examination in the second round. Additionally, land types with mean scores close to 3.5, specifically E4, E10, E11, S14, and C15, will also require careful evaluation for consensus in the next round. Consequently, the summary report after the first round was revised to provide detailed information on these land types, aiming to ensure objective expert feedback in the subsequent round.

4.2. Round 2

Following the analysis of the results from the first Delphi round, a second survey instrument, accompanied by a comprehensive summary of the initial round's findings, was distributed to the expert panel. Sixty experts participated in this second round, representing an 89.5% retention rate. Experts were again asked to assess the relevance of the proposed land types using a 5-point Likert scale. The questionnaire design mirrored that of the first round to facilitate the evaluation of consensus and consistency in expert opinions across both iterations.

Upon receiving the feedback, a detailed analysis of consensus indices among the experts was conducted. Land types identified for inclusion in the final framework for NLUP were those that achieved a clear consensus and exhibited an average score of 3.5 or higher, signifying a high priority level for national-level consideration.

Figure 6 illustrates the outcomes of the second Delphi round. This round demonstrated a high level of consensus among experts for nearly all land types across the five categories.

Economic development: Experts reached a consensus on 10 important land types for NLUP (Figure 6a). Initial disagreements observed in Round 1 concerning the importance level of land types for other annual crops (E3) and perennial crops (E4) for NLUP were largely resolved. By the conclusion of Round 2, both E3 and E4 were prioritized at a medium level with a high degree of consensus (IQR=1 and S.d <1).

Social Development: Similarly, all the 18 land types within this category garnered consensus from the expert panel. Notably, paddy land (S2) and land for infrastructure development (S12) emerged as the most significant land types in this group. While other land types, such as land for natural forests (S7), other annual crops (S3), salt production (S8), urban residential areas (S14), rural residential areas (S15), and land for religious establishments (S16), were less acknowledged for their social significance, as their prioritization remained at a medium level (with a mean score of less than 3.5). Consequently, these land types were not retained in the final list (Figure 6b).

Landscape Ecology-Environment: After Round 2, this category retained 9 land types, following the exclusion of lagoon (L9) and coastal surface water land (L11). Conversely, land with rivers, streams and special-use water surface (L7) achieved a higher mean score (mean = 4.08) and strong consensus (S.d=0.91 and IQR=1), thereby securing its retention in the refined list (Figure 6c).

Climate Change: A significant improvement in consensus and data reliability was observed for land types within the category of climate change. Following two rounds, the standard deviation values for desertification-prone land (C13) and flood-prone land (C15) were 1.01 and 1.04, respectively (Figure 6d). This minor deviation, coupled with a coefficient of variation (CV) difference less than 0.15 across the two rounds, indicated consistent expert opinion against the selection of these two land types for national-level planning. The remaining land types in this group include land at risk of flooding due to rising sea level (C9), salinity intrusion-prone land (C10), drought-prone land (C11), landslide-prone land (C12), special-use forest land (C3), protective forest land (C2), and productive forest land (C4).

Regional Linkage: A consistent trend was evident in the category of regional linkage, where 13 land types related to regional connectivity were prioritized with high consensus for inclusion in the NLUP after Round 2 (Figure 6e).

Accepted

Research

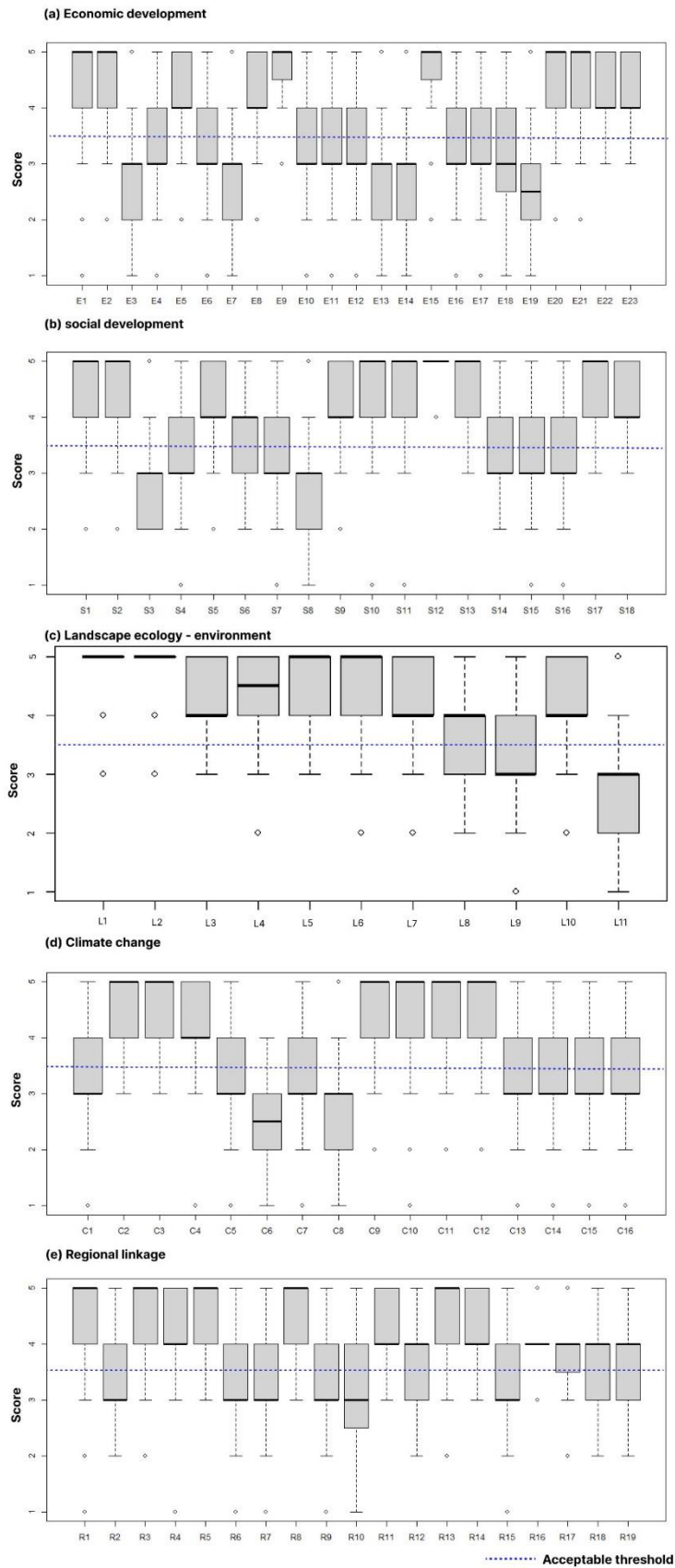


Figure 6. The Delphi results of Round 2

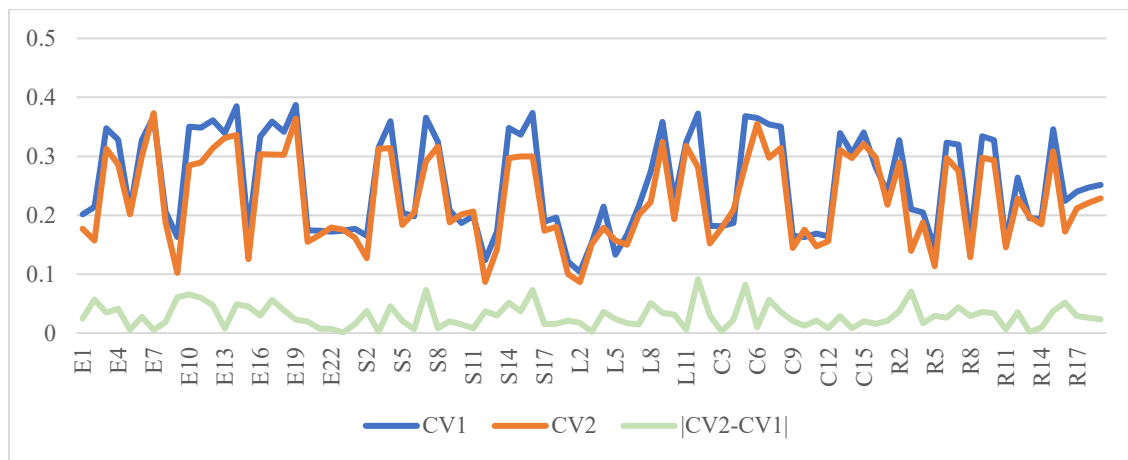


Figure 7. The variation coefficient of land types over the two rounds

As in previous studies, the Delphi survey process was concluded not only upon achieving expert consensus but also once the stability of responses was ensured, indicating statistical consistency for the same variable across survey rounds (Dajani et al., 1979). To ascertain dataset stability, the coefficient of variation (CV) was calculated using the mean and standard deviation values of land types. As illustrated in Figure 7, the largest variation recorded was 0.09 for C1 (rice cultivation land). While some experts believed that rice cultivation's capacity for carbon absorption and its potential for adaptation of climate change, the majority did not rank it as highly as forest land in this context. With all coefficients of variation values below 0.15, the Delphi survey was formally concluded.

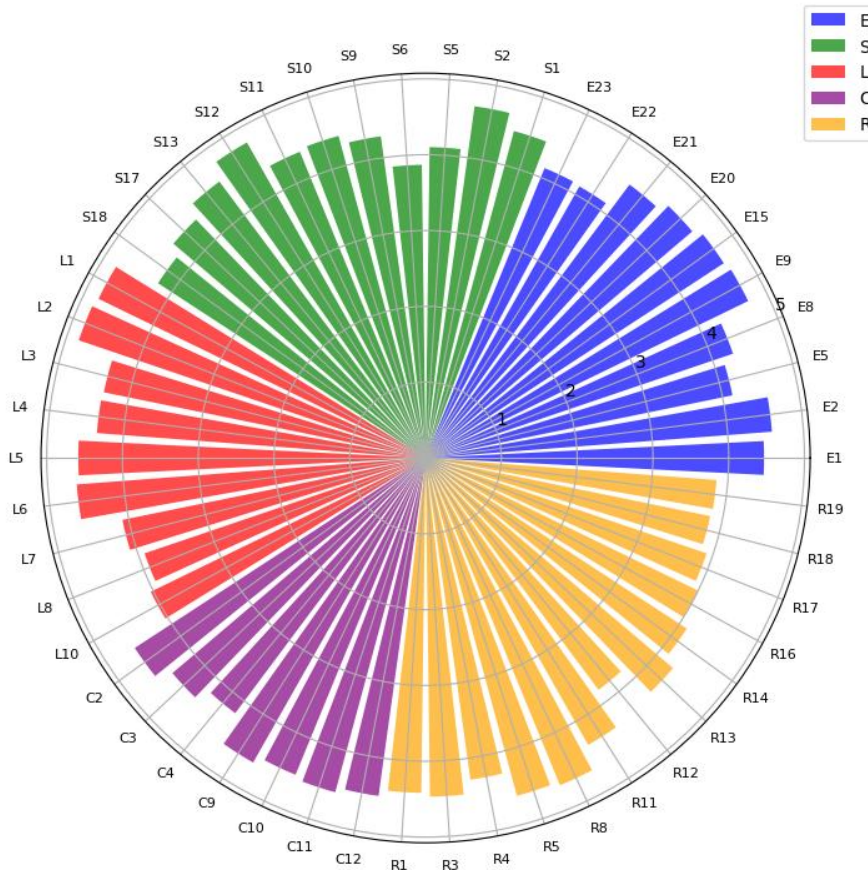


Figure 8. A proposed framework of land types for national land use planning in Vietnam with mean values

(Note: **E**: Economic development; **S**: Social development; **L**: Ecological landscape – Environment; **C**: Climate change; **R**: Regional Linkage)

Through these two rounds of the Delphi process, a robust set of 50 land types was established for the proposed framework, distributed across five integrated categories as follows: Economic development (10 indicators); Social development (11 indicators); Ecological landscape - Environment (9 indicators); Climate change (7 indicators), and Regional linkage (13 indicators) (Figure 8).

5. Discussion

5.1. The rationality of the proposed framework of land types

Traditional NLUP frameworks frequently lack explicit consideration of multi-dimensional factors, thereby limiting their capacity to comprehensively represent modern LUP perspectives. To address this deficiency within Vietnam's existing NLUP system, the proposed framework of land types systematically integrates crucial dimensional considerations. This approach directly aligns with the strategic and directional guidelines set forth by the Vietnamese government for its new

development phase (The Government of the Socialist Republic of Vietnam, 2020; The National Assembly of the Socialist Republic of Vietnam, 2017).

From a practical standpoint, the integration of these dimensions (Economic development, Social development, Landscape ecology – Environment, Climate Change, and Regional Linkage) facilitates a more targeted and effective response to Vietnam's pressing development challenges. While Vietnam has indeed achieved substantial economic growth within the Southeast Asian region, marked by a significant increase in GDP from 446 trillion VND in 1986 to 9,513 trillion VND in 2022, persistent issues continue to impede long-term sustainable development. These include limitations in resource mobilization, allocation, and utilization, alongside slow and inefficient regional and local linkages (Nguyen Tan Vinh & Than Thi Sen, 2023). Furthermore, the Vietnam Biodiversity Assessment Report (2021) highlighted ongoing degradation of the nation's ecological system and biodiversity, with "land use conversion pos[ing] a significant threat to Vietnam's rich biodiversity (Thuairé et al., 2021). Concurrently, Vietnam's consistent vulnerability to climate change is widely acknowledged (Espagne et al., 2021, p.10).

By systematically integrating these dimensional considerations, the proposed framework of land types enhances the state's capacity to efficiently and rationally distribute land types to regional jurisdictions. This allows for a tailored approach that considers their specific assets and developmental obstacles. Crucially, this framework serves as a foundational platform for subsequent, more detailed planning phases, which will involve assessing the relative importance of each land type across relevant categories and determining optimal land use areas and spatial configurations.

A significant obstacle to the effective implementation of NLUP traditionally stems from divergent political perspectives across governmental tiers (Jeronimo et al., 2015; Waite, 1970). In Vietnam, prior LUP initiatives have often been hampered by a lack of inter-level consistency, leading to discrepancies between national objectives and local demands (Ministry of Natural Resources and Environment, 2022, 63 pages: p.30). To mitigate these inherent conflicts and foster greater coherence, the NLUP framework must comprehensively encompass critical land types. Therefore, this proposed framework expands upon previous iterations, incorporating a broader range of land types derived from expert consensus and precisely aligned with the government's comprehensive development vision for the new period.

5.2. Recommendations for the integration of proposed framework of land types within Vietnam's NLUP system

Effective integration of climate change adaptation and mitigation into planning frameworks requires both horizontal and vertical approaches (Beck et al., 2009). Horizontal integration entails embedding necessary content into government documents and legal regulations, while vertical integration involves incorporating the relevant content into sectoral policies and their implementation procedures. This dual integration is essential during decision-making processes, the development of strategic plans at the ministerial level, and the formulation of strategies and actions at lower administrative levels.

To successfully implement the proposed framework of land types into Vietnam's NLUP system, the first and most critical step is to formally integrate dimensions related to landscape ecology, regional linkage, and climate change into the legal provisions governing the content of NLUP. This can be achieved through amendments to key legislation such as the Planning Law and the Land Law, as well as modifications to the procedures for formulating and adjusting NLUP, as stipulated by the Ministry of Natural Resources and Environment. Furthermore, the ministry's forthcoming guiding circulars, which will accompany the Land Law 2024, should clearly define and officially adopt the land types for NLUP based on the comprehensive framework developed by our research group.

6. Conclusion

In the current and forthcoming decades, Vietnam faces substantial challenges to land use planning stemming from intensified industrialization, rapid urbanization, and the pervasive impacts of climate change. Amidst these pressures, the integration of landscape ecology-environment, regional linkage, and climate change considerations into NLUP is not merely beneficial but crucial. Such integration is essential to ensure robust regional connectivity in land use, prevent land degradation and wastage, safeguard vital landscapes and natural ecosystems, and foster effective adaptation and response strategies to climate change. This paper has successfully addressed this imperative by developing a comprehensive framework comprising 50 distinct land types, systematically organized across five integrated categories.

Moving forward, Vietnam should aspire to achieve a higher level of sophistication in its land use planning practices. The practical application of the proposed framework of land types, with its inherent multi-dimensional focus, offers a tangible pathway towards realizing feasible and truly integrated land use planning. Furthermore, this framework is designed to serve as a robust foundation for the subsequent development of more granular regional and local land use plans. Acknowledging

that developing countries often encounter extensive difficulties and challenges in implementing this advanced planning (Delphin et al., 2022), it is paramount to recognize that the ultimate success of LUP extends beyond methodological advancement. It is also inherently dependent on the adaptive capacity of the government and the transparent and equitable enforcement of legal frameworks (USAID, 2021). These institutional and governance aspects represent critical challenges that Vietnam must effectively navigate in the future to translate NLUP from theoretical frameworks into practical, on-the-ground realities.

Acknowledgements: This research was funded by the National project “Research and develop a set of dimensions and indicators integrating landscape ecological factors, regional connectivity and climate change in land use planning to ensure sustainable development in Vietnam”, grant number: DTDLCN -94/21.

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